Hang On

AI call screening | Hackathon submission

Alexander Menassa | Baiwei Zhong | Boson AI × MScAC Hackathon | October 2026

01 Project information

Project name

Hang On

Track

Voice interaction

One-sentence description (showcase)

Hang On is an AI call screener that answers unknown calls in a persona tailored to the person it protects, draws scammers into revealing themselves, ends confirmed scams with evidence for the family, and puts ordinary callers through.

Write-up (195 words)

Problem. Scammers pose as banks, tax agencies and grandchildren, targeting older people who must judge in seconds whether a stranger is real. Call blockers only see numbers, not what is said.

Who it's for. Older adults and others likely to be targeted, and the families who worry about them.

How it works. Every unknown call reaches a gateway first. Blocked numbers are rejected, trusted numbers ring straight through, and everyone else is answered, voiced by Higgs Realtime, by a persona built to match the person being protected. Its name, age, backstory and voice are configurable; our demo uses Walter, an 88-year-old man. Scammers relax when they think they have reached the real person and make their real pitch, which is what we need to hear.

After each turn, a separate classifier (Gemini) reports scam behaviours such as gift-card payment, arrest threats or requests for codes, each backed by the caller's exact words. Plain code, not either model, decides. A scam ends the call with a polite excuse, blocks the number and writes an evidence card. Ordinary callers are put through. Bank and government callers are never put through automatically.

Phone numbers and SMS are simulated.

02 Screenshots

The browser demo, running on a phone and a laptop. All phone numbers are simulated and audio is browser audio only (the phone footers read "Browser audio, No carrier call").

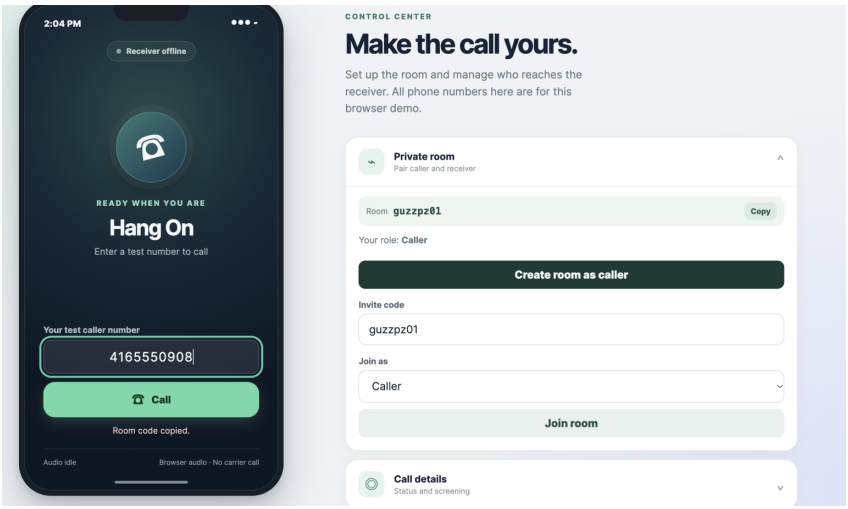


Figure 1. Caller view: creating a private room

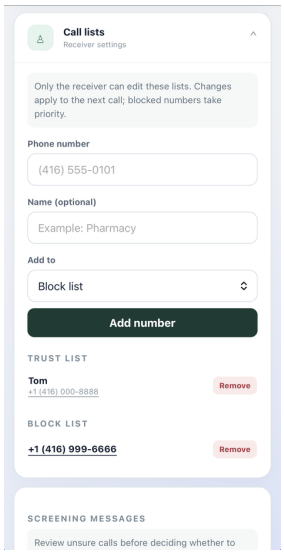


Figure 2. Receiver's trust and block lists

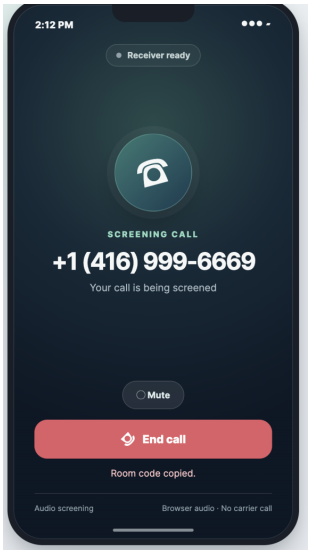


Figure 3. Unlisted number is screened

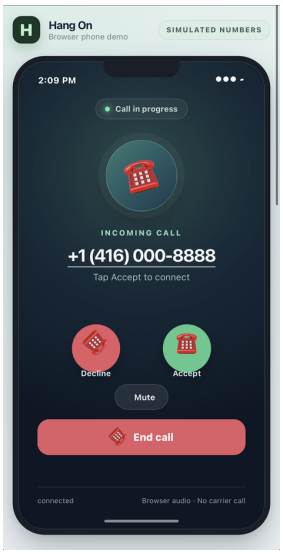


Figure 4. Trusted number rings the receiver

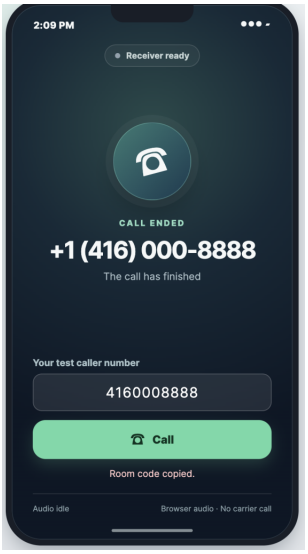


Figure 5. Call ended

- **Figure 1, caller and room setup.** The caller creates a private room and shares its code with the receiver. The phone-style screen shows "Receiver offline" until the receiver joins.
- **Figure 2, receiver's call lists.** Only the receiver can edit the trust and block lists. Changes apply to the next call, and blocked numbers take priority. Shown: a trusted contact, Tom, and a blocked number.
- **Figure 3, unknown number screened.** The number +1 (416) 999-6669 is on neither list (it differs by one digit from the blocked +1 (416) 999-6666), so Hang On screens the call ("Your call is being screened") instead of ringing the receiver.
- **Figure 4, trusted number rings the receiver.** Tom's number rings the receiver directly, with Accept and Decline, and no screening.
- **Figure 5, call ended.** The caller's view after the trusted call finishes.

03 Repository

Repository URL

`github.com/AlexanderMenassa/BosonAI-Hackathon` public repository, branch `main`

What the judges will find

- Source code in `hangon/`: decision engine, classifier, Higgs voice and text agents, call gateway, browser demo.
- Setup and run instructions in `README.md`, including three real example inputs and outputs (a scam call, an ordinary caller, and a repeat caller who is rejected).
- Scripted example calls in `scenarios/`; over 200 offline tests in `tests/`.
- `.env.example` lists every setting by name, with no values. No credentials are committed.

Quick start for judges (no API keys needed)

```
python3 -m venv .venv && source .venv/bin/activate
pip install -r requirements.txt
python -m pytest -q
python -m hangon.sim --scenario scenarios/cra_scam.txt --number
+14165550111
```

04 How you built it

Technical write-up (190 words)

Higgs Realtime (`higgs-realtime`) is the persona's live voice over a server-side WebSocket. We turn automatic replies off (`create_response: false`) so our code decides each turn before the model speaks, and send each reply's role as per-reply instructions. When the caller talks over the persona, we flush playback and truncate the stored reply (`conversation.item.truncate`) to what was actually heard.

Higgs STT (`higgs-stt-3.1`) transcribes each caller turn inside the realtime session and feeds the classifier.

Boson Voices API registers any reference clip from the hackathon library as the persona's custom voice, automatically at startup under each user's own key, so the voice can match any user.

Hard and novel:

- *Separated authority:* Higgs speaks, Gemini assesses, code decides, so a caller cannot talk the agent into a transfer.
- *Grounded evidence:* every behaviour the classifier reports must quote the caller verbatim; unmatched quotes are dropped, so it cannot invent a scam.
- *No dead air:* classification runs in the background. A late verdict is applied after the current line, and a stale "legit" result can never forward a caller whose newer turn sounded like a scam.
- A Gemini-driven red-team harness plays scammers from written briefs.

05 Team information

Name	GitHub	Worked on
Alexander Menassa	@AlexanderMenassa	Backend: all of the logic behind the call, including the decision engine, the judge and classifier, the persona, and the voices.
Baiwei Zhong	@vito-zbw	Frontend and the virtual calling system, including the messages.

Known limitations and things judges should know

- Phone numbers, the telephone network and SMS or guardian alerts are **simulated**. Calls enter through the browser or terminal, not a real phone line, and caller ID can be spoofed, so a trusted number is not identity verification.
- Evaluated only on a small synthetic set of scripted and red-team calls. This is not evidence of real-world scam-detection accuracy; false positives and negatives are possible.
- With Gemini, call transcripts leave the machine; an Ollama option keeps them local.
- Background classification is wired into the terminal voice call and simulator. The browser demo still waits for the classifier before Walter replies.
- The browser demo was tested live on two devices, a phone and a laptop. Networks that block direct WebRTC connections may need a TURN relay, which is supported but must be configured (see `docs/PHASE-2B-SETUP.md`).
- English-first; other languages are untested. An evidence-logging tool via Higgs function calling was planned but not built.

Looking ahead

Hang On proves the core loop: answer, listen, judge, decide. These are the next steps toward something a family could rely on.

- **Answer in the protected person's own voice.** Our demo persona, Walter, is a stand-in. With the protected person's consent, voice mimicry could copy their actual voice, so callers hear who they expect and a handoff to the real person sounds seamless. The pipeline for registering a custom voice from a short reference clip already exists.
- **Know the person, catch scams faster.** Add personal details, such as their bank, pharmacy, doctor and family members' names, and check callers' claims against them. A "grandson" nobody has heard of, or a "bank" they do not use, can be flagged sooner. The details would stay private and never be revealed to a caller.
- **A fast model and a real phone number.** Run classification on a faster model to shorten the delay, and connect Hang On to a real phone number through a telephony provider. Proper testing needs real calls, with real latency, audio quality and accents, which a browser demo cannot reproduce.
- **Automatic trusted and blocked lists.** Import the phone's contacts as trusted numbers, with permission, and fill the blocklist from known scam numbers, instead of managing lists by hand. Caller ID can be spoofed, so a trusted number should still never be put through for high-stakes requests without checks.
- **Guardian check-ups.** The system already sends the same summaries to an optional guardian, so a family member can check on the vulnerable person. In this demo those messages are simulated, and we did not demo or test them rigorously. Next: real delivery and thorough testing.
- **Bug fixes and hardening.** Fix known issues, for example making the browser demo classify in the background like the voice call so the assistant never pauses, and test more scam types, languages and live calls.